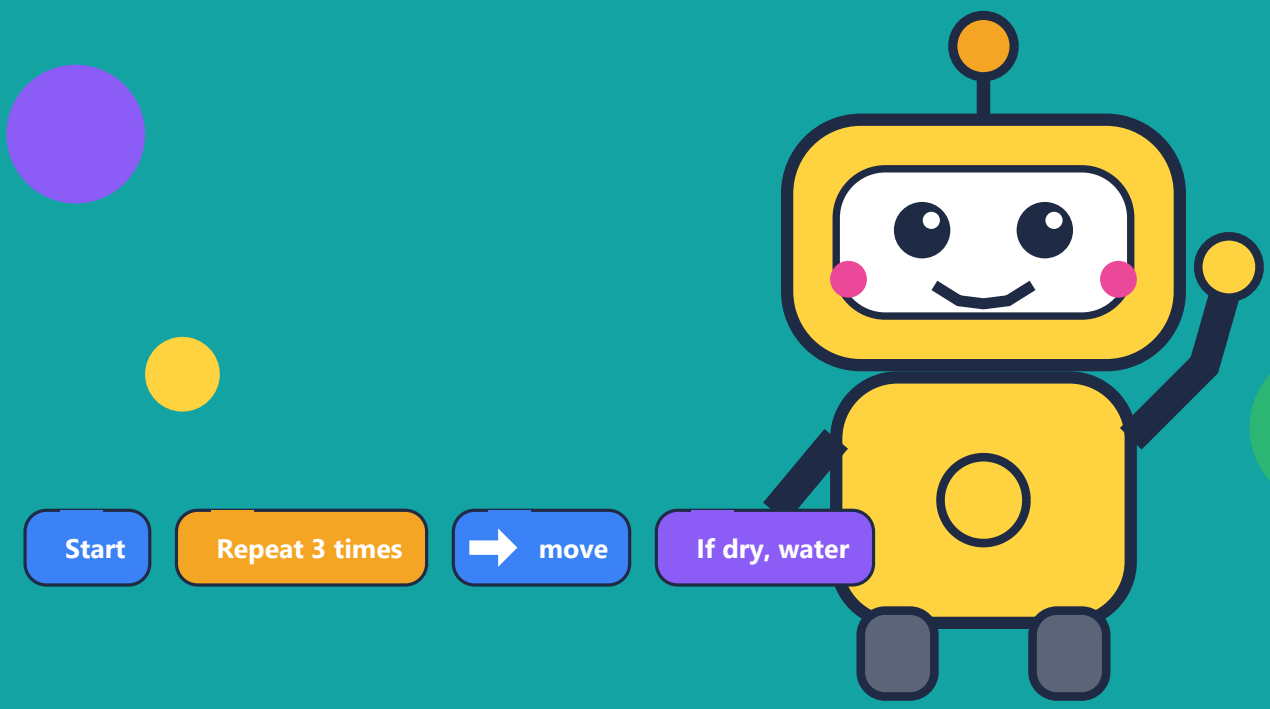
CodeSchool · Class 4

Think Like a Coder

Computational thinking for Class 4

Learn it · See it · Try it · Build it



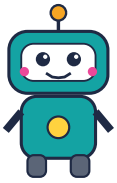
Follows the 14 chapters of the NCERT Class 4 maths textbook

Built for CBSE's Computational Thinking and AI curriculum, 2026-27

Theory, worked examples, practice, coding on paper and challenges

SAMPLE EDITION · DRAFT

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Hello! I am Robo. I think like a coder, and soon you will too! In this book we will solve puzzles, crack secret codes, guide robots and write programs, all on paper.

How this book works

Learn — Robo explains one big idea in simple words, with the new coder words to remember.

Example — A worked example shows the idea step by step.

Try it! — Five practice questions. Circle the answer or write it in the box.

Now you build it — Coding on paper: put code cards in order, fix bugs and write your own programs.

Challenge — A harder puzzle for when you are ready for more.

How did I do? — Colour a face to show how sure you feel.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this | **Repeat: a loop** | **If: a decision** | **When: an event** | **move**

For teachers and parents

Each chapter goes with the same chapter of the NCERT Class 4 maths textbook, so it can be used in maths periods. CBSE suggests about 50 hours of computational thinking a year in Classes 3-5. The coding here needs no computer: children act out, write and check programs on paper, which builds the same ideas used later in block coding and AI. Answers are at the back. Encourage children to explain their thinking, not just the answer.

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Coder skills in this book

Levels: B1-B2 reviewed; B3 if-else decisions and tracing; B4 sorting networks, nested and repeat-until loops.

Ch	Coder skill	Level	Ch	Coder skill	Level
1	Faces, edges and nets	B1 review	8	If-else decisions and measuring steps	B3
2	If-then-else choices	B3	9	Loops inside loops	B4
3	Odd, even and checking for mistakes	B3	10	Clue tables and data	B3
4	Big numbers and sorting machines	B4	11	Mirror patterns and repeat with turns	B4
5	Fractions and checking a condition	B3	12	Repeat until and cycles	B4
6	Shortest paths and loops with turns	B4	13	Decision trees: sorting by questions	B3
7	Tracing a machine step by step	B3	14	Collecting and showing data	B2 review

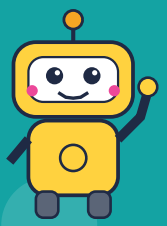
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CHAPTER 1 · LEVEL B1 review

1 Shapes Around Us

Coder skill: Faces, edges and nets

Goes with NCERT maths, Class 4, chapter 1

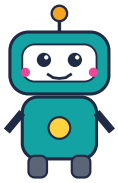


Learn

A **cube** has 6 flat **faces**, 12 straight **edges** and 8 **corners**. A cuboid, like a matchbox, has the same numbers.

If you cut a box open and flatten it, you get a **net**. A net is a flat shape that folds into a solid. Turning a 3D object into a flat plan keeps only what matters: that is **abstraction**.

face	edge	net
a flat side of a solid	where two faces meet	a flat shape that folds into a solid

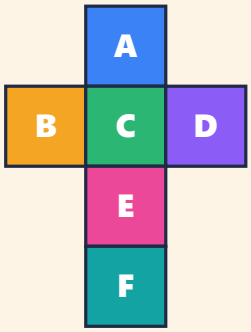


A box is just six flat faces folded up. Coders call the flat pattern a net. Seeing a solid as a net is abstraction!

Example

Which faces are opposite?

1. Fold the net: C is the bottom, A, B, D and E stand up around it, and F folds over to make the top.
2. In the long strip A, C, E, F, faces that are one apart are opposite: **A and E**, **C and F**.
3. The two side faces **B and D** are also opposite.



★ *Coder fact!* Cardboard boxes are cut as flat nets and folded by machines that follow programs.

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Try it!

Circle the right answer, or write it in the box.

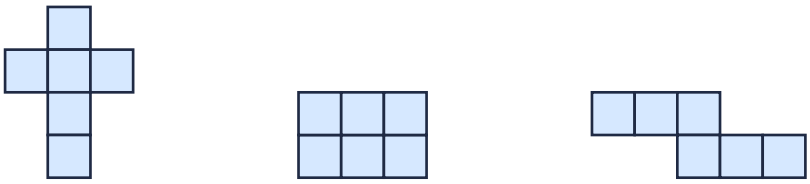
1 How many faces, edges and corners does a cube have?

☐ ☐ ☐

2 Look at the net in the example. Which face is opposite **B**?

a) A **b)** C **c)** D **d)** F

3 Which of these will **NOT** fold into a cube?



a b c

a) a **b)** b **c)** c **d)** all fold

4 On a dice, opposite faces always add up to 7. If the top shows 2, what is on the bottom?

☐

5 Card: *Put 3 cubes in a row. Put 1 cube on top of the middle one.* Which picture matches?



a b c

a) a **b)** b **c)** c

Now you build it

Write drawing steps so a friend can draw a cube net on squared paper.

Draw 4 squares in a column | **Add a square on the left of the 2nd** | **Add a square on the right of the 2nd** | **Cut out and fold**

Follow the steps, cut out the net and fold it. Does it make a cube? Write which faces are opposite.

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★ Challenge

27 small cubes make a big 3 x 3 x 3 cube. How many small cubes can you NOT see from any side, top or bottom?

✓ How did I do?

I can count faces, edges and corners.	I can find opposite faces in a net.	I can follow drawing steps exactly.
		

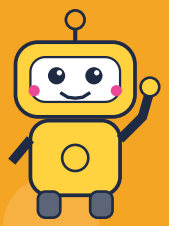
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CHAPTER 2 · LEVEL B3

2 Hide and Seek

Coder skill: If-then-else choices

Goes with NCERT maths, Class 4, chapter 2

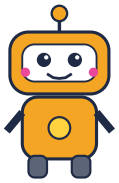


Learn

Computers make choices with **if-then-else**: *if* something is true, *then* do one thing, *else* do another.

When an if-else sits **inside a loop**, it is checked again every time the loop repeats. That is how a robot reacts to each new square.

if-then-else	condition
a choice between two actions	the question that is true or false



I check my if-else rule on every square: if it is a puddle I say 'Splash!', else I say 'Step'. Two choices, every time!

Example

Robo walks along the path

1. Program: *repeat 4 times: move forward; if on a puddle (grey) then say 'Splash!' else say 'Step'.*
2. Robo lands on tiles 1, 2, 3, 4: grey, white, grey, white.
3. Robo says: **Splash, Step, Splash, Step**.



★ *Coder fact!* A lift uses if-then rules: if someone presses a floor button, then it stops at that floor.

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Try it!

Circle the right answer, or write it in the box.

1 Use the same program on this path. What does Robo say?



2 Rule: *if the number is more than 10 then clap, else jump*. What do you do for 7, 12 and 10?

3 How many times does Robo say 'Splash!' on this path?

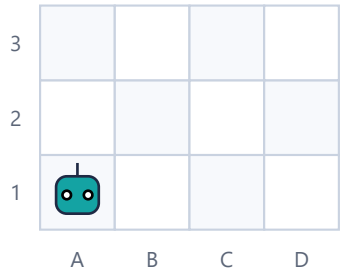


a) 2 **b)** 3 **c)** 4 **d)** 6

4 Robo is hiding in the red, blue or green box. *Robo is not in the red box. If Robo were in the green box, it would be open. The green box is closed.* Where is Robo?

a) red **b)** blue **c)** green **d)** can't tell

5 Robo is on A1 facing **up**. Program: *forward, turn right, forward, forward*. Which square does Robo reach?



a) C2 **b)** B3 **c)** A3 **d)** C1

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Now you build it

Write a program that makes Robo say 'Hot!' on sun tiles and 'Cold!' on snow tiles along a path of 5 tiles.

Repeat 5 times | move forward | If on a sun tile | say 'Hot!' | else | say 'Cold!'

Draw a path of 5 sun and snow tiles. Swap with a friend and trace each other's program.

Challenge

On a path of 10 tiles, every 3rd tile is a puddle. Robo runs the 'Splash or Step' program on all 10 tiles. How many times does it say 'Splash!'?

How did I do?

I can follow an if-then-else rule.	I can trace a program with a choice.	I can write my own if-else program.
		

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3

CHAPTER 3 · LEVEL B3

Patterns Around Us

Coder skill: Odd, even and checking for mistakes

Goes with NCERT maths, Class 4, chapter 3



Learn

An **even** number can be shared into pairs with nothing left over. An **odd** number always has one left over.

Computers use this to catch mistakes. They add an extra **check** to each row so the row has an even number of 1s. If one value flips, that row becomes odd, and the computer knows something went wrong. This is called a **parity check**.

even	odd	parity check
shares into pairs	one left over	an even/odd check for mistakes

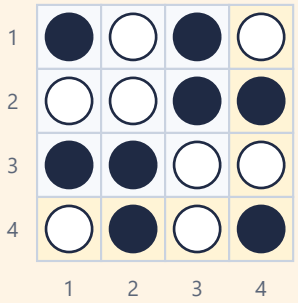


Computers add a check to every row of data. If one thing flips, the row turns odd, and the computer spots the mistake!

Example

Coin Flip Magic

1. In this grid every row and every column has an **even** number of black coins. The yellow row and column are the check coins.
2. A friend secretly flips one coin. Now one row and one column have an **odd** number of black coins.
3. Where that row and column cross is the flipped coin.



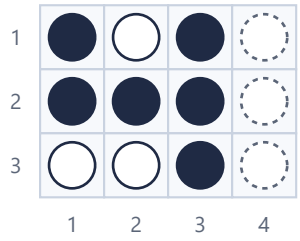
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Try it!

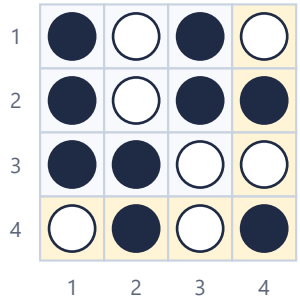
Circle the right answer, or write it in the box.

1 Circle the odd numbers: 14, 27, 30, 41, 58.

2 Each row needs an even number of black coins. What colour must the empty check coin be in each row?



3 One coin in the example grid was flipped, and now it looks like this. Which coin was flipped?



a) row 1, column 1 **b)** row 2, column 1 **c)** row 2, column 3 **d)** row 3, column 2

4 Odd + odd is always...

a) even **b)** odd **c)** sometimes even

5 What comes next: **2, 4, 8, 16, ?**

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Now you build it

Put the Coin Flip Magic steps in order, then perform the trick for a friend.

Find the row and the column that are odd

Lay out a 4 x 4 grid of coins

Friend flips one coin

Add a check coin to make each row and column even

Number the cards 1 to 4, then try the trick at home with buttons or paper circles.

Challenge

If a friend flips TWO coins in the same row, can you still find them? Explain.

How did I do?

I can tell odd from even.	I can add a check coin.	I can find a flipped coin.
		

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CHAPTER 4 · LEVEL B4

4 Thousands Around Us

Coder skill: Big numbers and sorting machines

Goes with NCERT maths, Class 4, chapter 4

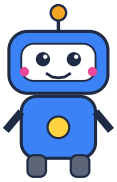


Learn

Big numbers have a **thousands** place: 4,072 means 4 thousands, 0 hundreds, 7 tens and 2 ones.

A **sorting network** sorts numbers with **comparators**. Two numbers enter a comparator; the smaller goes to the upper wire and the bigger to the lower wire. Computers can do many comparisons at the same time, so they sort very fast.

thousands the 4th place from the right	**comparator** compares two numbers and orders them	**sorting** putting in order

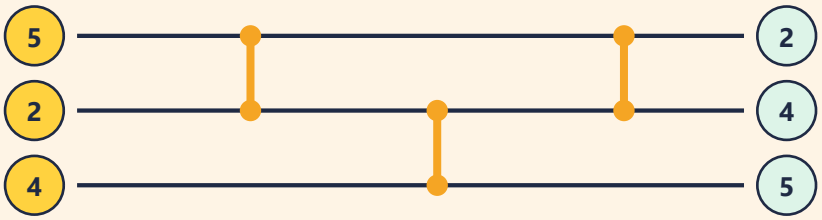


In a sorting network, numbers race along wires. At each comparator the smaller one goes up. At the end they come out in order!

Example

Sort 5, 2, 4 with the network

1. Comparator 1 (top two wires): 5 and 2, the smaller goes up: **2, 5, 4**.
2. Comparator 2 (bottom two wires): 5 and 4: **2, 4, 5**.
3. Comparator 3 (top two wires): 2 and 4 stay. Out come **2, 4, 5**, smallest on top.



★ *Coder fact!* Big websites sort millions of items in a blink by doing many comparisons at the same time.

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Try it!

Circle the right answer, or write it in the box.

1 Write 4,072 in parts: ___ thousands, ___ hundreds, ___ tens, ___ ones.

2 Order from smallest to largest: 3,410; 3,041; 3,401; 3,140.

3 7 and 3 enter a comparator. Which number comes out on the upper wire?



a) 7 **b)** 3

4 Send 9, 4, 6 through the same network as the example. What comes out, top to bottom?



5 Use the digits 5, 0, 8 and 1 once each. What is the largest number you can make?

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Now you build it

Turn your class into a sorting network for 3 friends of different heights.

Friends stand on wires 1, 2, 3

Compare wires 1 and 2: shorter goes to 1

Compare wires 2 and 3: shorter goes to 2

Compare wires 1 and 2 again

Try it with three friends. Do they always end up shortest to tallest?

Challenge

Could just the first two comparators always sort 3 numbers? Try 3, 2, 1.

How did I do?

I can read numbers in the thousands.	I can order big numbers.	I can use a sorting network.
		

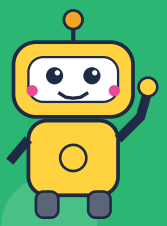
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CHAPTER 5 · LEVEL B3

5 Sharing and Measuring

Coder skill: Fractions and checking a condition

Goes with NCERT maths, Class 4, chapter 5

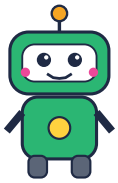


Learn

A **fraction** names equal parts of a whole. If a roti is cut into 4 equal parts, each part is **one quarter** (1/4). Two quarters make one half.

Before we act, we can check a **condition**: *if* all the parts are equal, *then* the share is fair, *else* it is not.

fraction	quarter	condition
equal parts of a whole	one of 4 equal parts	something we check before acting

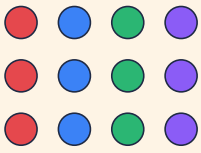


Before I say a share is fair, I check a condition: are all the parts equal? If yes, then fair. Else, not fair!

Example

One quarter of 12 laddoos

1. Share 12 laddoos into 4 equal groups.
2. Each group has **3** laddoos.
3. So one quarter of 12 is **3**, and three quarters of 12 is 9.



★ *Coder fact!* Recipe apps use fractions: halve a recipe and every amount becomes one half.

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Try it!

Circle the right answer, or write it in the box.

1 What fraction of the strip is shaded?



a) 3/5 **b)** 3/8 **c)** 5/8 **d)** 1/3

2 Which is more of the same roti: one half or one quarter?

a) one half **b)** one quarter **c)** same

3 Rule: *if a number shares into 2 equal parts with nothing left, then it is even.* Is 15 even?

a) yes **b)** no

4 Pens cost the same and Riya spends exactly Rs 20. Which can NOT be the price of one pen?

a) Rs 4 **b)** Rs 5 **c)** Rs 6 **d)** Rs 10

5 A jug holds 1 litre. How many 250 ml glasses can it fill?

Now you build it

Write a checking program that decides if a share is fair.

Count the pieces each child gets | If every child has the same number | say 'Fair!' | else | say 'Not fair!'

Test it: 10 sweets shared as 4, 3, 3. What does the program say?

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★ Challenge

What is three quarters of 20? Show it with a drawing.

✓ How did I do?

I can name fractions.	I can compare halves and quarters.	I can check a condition before deciding.
		

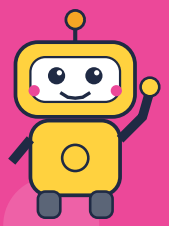
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CHAPTER 6 · LEVEL B4

6 Measuring Length

Coder skill: Shortest paths and loops with turns

Goes with NCERT maths, Class 4, chapter 6

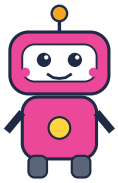


Learn

We measure length in **centimetres** and **metres**: 1 m = 100 cm. The **shortest path** between two places uses the fewest steps.

Now Robo can **turn left** or **turn right** from where it is facing, then move **forward**. A loop with turns can draw shapes: *repeat 4 times: forward 3, turn right* draws a square.

centimetre	turn	shortest path
100 make 1 metre	change the way Robo faces	the route with fewest steps

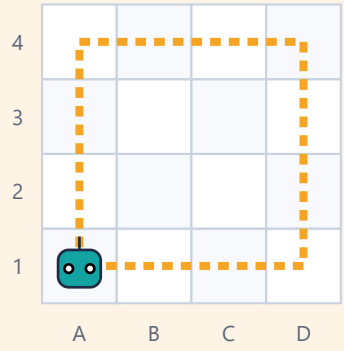


Repeat 4 times: forward 3, turn right. That little loop draws a whole square!

Example

Drawing a square with a loop

1. Robo starts at A1 facing up.
2. Repeat 4 times: forward 3, turn right.
3. Robo goes up 3, right 3, down 3, left 3, and ends where it started: a **square**.



★ *Coder fact!* Map apps find the shortest route by comparing many possible paths very quickly.

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Try it!

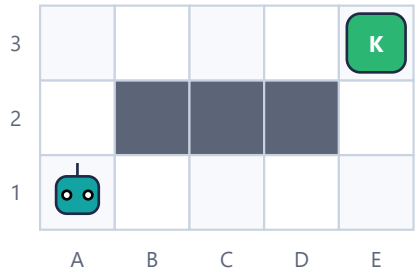
Circle the right answer, or write it in the box.

1 How many centimetres are there in 2 metres?

2 What shape does *repeat 4 times: forward 2, turn right* draw?

a) triangle **b)** square **c)** line **d)** circle

3 Robo is on A1. The grey squares are walls. What is the fewest number of moves to reach the kite K on E3?



4 Which is longer: 2 m or 150 cm?

a) 2 m **b)** 150 cm **c)** same

5 Robo faces up. It turns right, then turns right again. Which way does it face now?

a) up **b)** down **c)** left **d)** right

Now you build it

Write a program that draws a rectangle 3 squares long and 2 squares tall.

Repeat 2 times | forward 3 | turn right | forward 2 | turn right

Walk it on the floor grid or draw it on squared paper. Does Robo end where it started?

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★ Challenge

Does *repeat 3 times: forward 4, turn right* make a closed shape? Why or why not?

✓ How did I do?

I can change metres to centimetres.	I can find a shortest path.	I can draw shapes with a loop and turns.
		

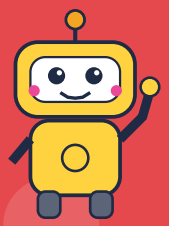
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CHAPTER 7 · LEVEL B3

7 The Cleanest Village

Coder skill: Tracing a machine step by step

Goes with NCERT maths, Class 4, chapter 7

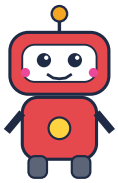


Learn

Many machines have a **state**: how things are right now, like a lamp being ON or OFF.

Pressing a button changes the state. To know the final state, we **trace**: write down the state after every single press, in order.

state	toggle	trace
how things are right now	switch ON to OFF or OFF to ON	follow step by step



My lamps are either ON or OFF. That is their state. Each button press changes the state, so I trace one press at a time!

Example

Two lamps and two buttons

1. Both lamps start OFF. Button 1 flips lamp A. Button 2 flips both lamps.
2. Press 1: A ON, B OFF. Press 2: A OFF, B ON. Press 1: A ON, B ON.
3. After pressing 1, 2, 1 both lamps are **ON**.

Press	Lamp A	Lamp B
start	OFF	OFF
1	ON	OFF
2	OFF	ON
1	ON	ON

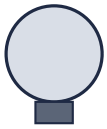
★ *Coder fact!* A washing machine moves through states: fill, wash, rinse and spin, changing state when each step ends.

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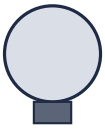
Try it!

Circle the right answer, or write it in the box.

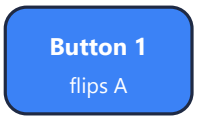
1 Both lamps start OFF. You press 2, then 2. What are the lamps now?



lamp A



lamp B





a) both ON **b)** both OFF **c)** A ON, B OFF **d)** A OFF, B ON

2 Both lamps start OFF. You press 1 three times. What are the lamps now?

a) A ON, B OFF **b)** A OFF, B OFF **c)** both ON **d)** A OFF, B ON

3 Both lamps are OFF. What is the smallest number of presses to turn both ON?

4 A cleaning robot follows: *if the bin is full, then empty it.* The bins are full, empty, full, full. How many bins does it empty?

5 Fill the trace table: lamps start OFF, presses 2, 1.

Press	Lamp A	Lamp B
start	OFF	OFF
2		
1		

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Now you build it

Design your own machine with two lamps and two buttons. Decide what each button does.

When button 1 is pressed — flip lamp ___

When button 2 is pressed — flip lamp ___

Give a friend three presses to trace. Check their final state.

Challenge

Both lamps start OFF. What is the fewest number of presses to get A OFF and B ON? Which presses?

How did I do?

I can describe a state.	I can trace a machine press by press.	I can design a simple machine.
		

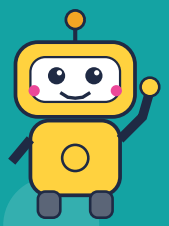
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CHAPTER 8 · LEVEL B3

8 Weigh It, Pour It

Coder skill: If-else decisions and measuring steps

Goes with NCERT maths, Class 4, chapter 8

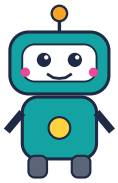


Learn

A balance helps us decide with **if-else**: *if* the left pan goes down, the left is heavier; *else* the right is heavier or they are equal.

With two jugs we can measure amounts we have no jug for, by **filling, pouring and emptying** in the right order. The list of pouring steps is an algorithm.

heavier weighs more	**litre** a unit for liquids	**kilogram** 1,000 grams

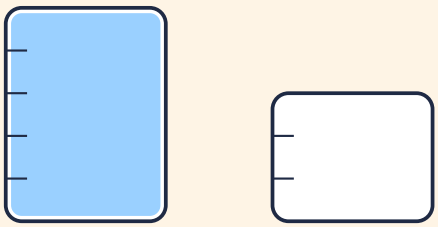


If the left pan goes down, the left side is heavier, else it is not. And to measure with jugs, I plan my pouring steps first!

Example

Measure 2 litres with a 5 L jug and a 3 L jug

1. Fill the 5 L jug.
2. Pour from the 5 L jug into the 3 L jug until the 3 L jug is full.
3. The 5 L jug now holds exactly **2 litres**.



5 L jug 3 L jug

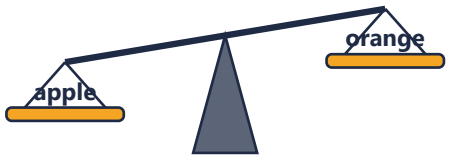
★ *Coder fact!* Post offices weigh parcels and use if-else rules to decide the price.

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Try it!

Circle the right answer, or write it in the box.

1 Which is heavier?



a) apple **b)** orange **c)** same

2 Rule: *if the bag weighs more than 5 kg, then carry it with two hands, else with one hand.* The bag weighs 7 kg. How do you carry it?

a) one hand **b)** two hands

3 How many grams make 1 kilogram?

4 Ball A is heavier than ball B. Ball C is lighter than ball B. Order them from heaviest to lightest.

5 The 5 L jug is full and the 3 L jug is empty. You pour from the 5 L jug until the 3 L jug is full, then empty the 3 L jug. How much water is left in total?

Now you build it

Write an if-else program for a parcel counter.

If the parcel weighs more than 2 kg **put it on the truck** **else** **put it on the bike**

Where do parcels of 1 kg, 3 kg and 2 kg go?

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★ Challenge

Using only a 5 L jug and a 3 L jug, write the steps to get exactly 4 litres.

✓ How did I do?

I can compare with a balance.	I can use an if-else rule.	I can plan pouring steps.
		

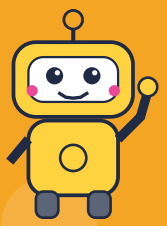
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CHAPTER 9 · LEVEL B4

9 Equal Groups

Coder skill: Loops inside loops

Goes with NCERT maths, Class 4, chapter 9

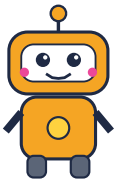


Learn

A **nested loop** is a loop inside another loop. The inner loop finishes all its repeats each time the outer loop runs once.

Nested loops make **arrays**: rows and columns. 3 rows of 4 is 3 x 4 = 12. Multiplication is a loop inside a loop!

nested loop	array	row
a loop inside a loop	objects in rows and columns	a line going across

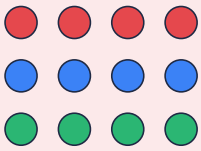


Repeat 3 times: [repeat 4 times: draw a star]. A loop inside a loop draws 3 rows of 4 stars: 12 stars!

Example

Stars in rows

1. Outer loop: repeat 3 times (one row each time).
2. Inner loop: repeat 4 times: draw a star. Then start a new row.
3. Total stars: 3 rows x 4 stars = **12**.



★ *Coder fact!* Screens are grids of tiny dots called pixels. A computer colours them row by row with loops inside loops.

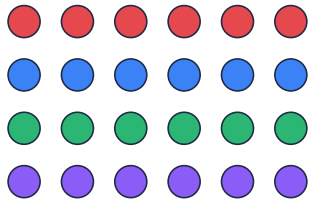
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Try it!

Circle the right answer, or write it in the box.

1 How many stars does this draw? *Repeat 5 times: [repeat 3 times: draw a star], new row.*

2 How many dots are in this array?



3 Which nested loop draws 18 stars?

a) repeat 3 x [repeat 6]
b) repeat 4 x [repeat 4]
c) repeat 5 x [repeat 3]
d) repeat 2 x [repeat 8]

4 There are 6 boxes with 5 pencils in each. How many pencils in all?

5 Draw what this makes: *repeat 2 times: [repeat 3 times: draw a circle], new row.*

Now you build it

Plan a garden with 4 rows and 5 plants in each row. Write it as a nested loop.

Repeat ___ times **Repeat ___ times** **plant a seed** **start a new row**

How many seeds do you need?

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★ Challenge

Repeat 3 times: [repeat 2 times: clap], stamp. How many claps and how many stamps?

✓ How did I do?

I can count with a nested loop.	I can read an array.	I can write a loop inside a loop.
		

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CHAPTER 10 · LEVEL B3

10

Elephants, Tigers and Leopards

Coder skill: Clue tables and data

Goes with NCERT maths, Class 4, chapter 10



Learn

A **clue table** (logic grid) keeps track of clues. Put a cross where something is not possible and a tick where it is.

When only one choice is left in a row, it must be true. Wildlife teams also keep **data** tables to count and compare animals.

clue table	data	logic
a grid to track clues	facts we collect and count	reasoning step by step



I use a clue table. A cross means 'no', a tick means 'yes'. When only one box is left in a row, it must be the answer!

Example

Who lives in which enclosure?

1. Clues: the elephant is not in enclosure 1. The tiger is in enclosure 3.
2. Tiger takes 3. The elephant can't be in 1, so it is in **2**.
3. Only enclosure **1** is left for the leopard.

Animal	Enclosure 1	Enclosure 2	Enclosure 3
Elephant	x	yes	x
Tiger	x	x	yes
Leopard	yes	x	x

★ *Coder fact!* Wildlife teams use camera traps and data tables to count tigers in Indian forests.

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Try it!

Circle the right answer, or write it in the box.

1 Asha, Ben and Chitra like red, blue and green (one each). Asha likes blue. Ben does not like green. Who likes what?

2 A park counted animals: elephants 12, tigers 5, leopards 8. How many animals in all?

3 From the same count, which animal was seen the least?

a) elephant **b)** tiger **c)** leopard

4 Clue: *I have stripes and I am a big cat.* Which animal am I?

a) leopard **b)** zebra **c)** tiger **d)** elephant

5 An elephant eats about 150 kg of food a day. About how much in 2 days?

Now you build it

Make a clue puzzle for a friend: 3 children, 3 favourite fruits. Write two clues that are enough to solve it.

Clue 1: ___ likes ___ **Clue 2: ___ does not like ___**

Can your friend solve it with a clue table? Are both clues needed?

Challenge

Elephant, tiger, leopard and deer live in cages 1-4. The elephant is in cage 4. The tiger is next to the elephant. The deer is not in cage 1. Where is each animal?

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✓ How did I do?

I can use a clue table.	I can read a data table.	I can write my own clue puzzle.
		

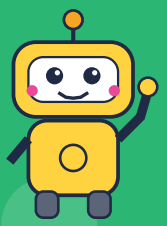
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CHAPTER 11 · LEVEL B4

11 Fun with Symmetry

Coder skill: Mirror patterns and repeat with turns

Goes with NCERT maths, Class 4, chapter 11

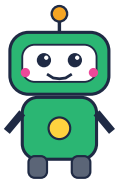


Learn

A shape has a **line of symmetry** if you can fold it so the two halves match exactly. Each part on one side has a **mirror** part at the same distance on the other side.

Loops with turns make symmetrical designs, like rangoli: *repeat 4 times: draw a petal, turn right.*

symmetry	line of symmetry	mirror image
both halves match	the fold line	a flipped copy

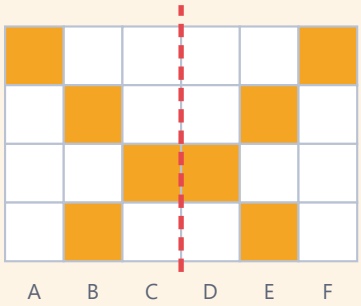


Repeat 4 times: draw a petal, turn right. A loop with turns makes a flower that looks the same all the way round!

Example

Complete the mirror pattern

1. The red dashed line is the mirror. A square that is 1 square from the line on the left...
2. ...has a partner 1 square from the line on the right, in the same row.
3. Copy each shaded square across the line to finish the pattern.



★ *Coder fact!* Rangoli designs often use repeat-and-turn symmetry, just like a loop with turns.

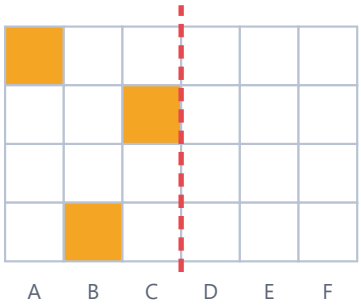
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Try it!

Circle the right answer, or write it in the box.

1 Which letters have a line of symmetry? A, F, H, J

2 Shade the right side to make the pattern symmetrical. Which column will the top-left square's partner be in?



a) D **b)** E **c)** F

3 How many lines of symmetry does a square have?

a) 1 **b)** 2 **c)** 4 **d)** 0

4 *Repeat 4 times: draw a heart, turn right.* How many hearts will there be?

5 An arrow turns a quarter turn each step: up, right, down, ... Which way next?



a) up **b)** left **c)** right

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Now you build it

Write a loop that draws a flower with 6 petals.

Repeat ___ times | draw a petal | turn a little to the right

Draw your flower. Does it have lines of symmetry? Mark them.

Challenge

A pattern on a 4-column grid (A, B, C, D) is symmetric about the line between B and C. Square C2 is shaded. Which other square must be shaded?

How did I do?

I can find lines of symmetry.	I can complete a mirror pattern.	I can draw a design with repeat and turn.
		

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12

CHAPTER 12 · LEVEL B4

Ticking Clocks and Turning Calendar

Coder skill: Repeat until and cycles

Goes with NCERT maths, Class 4, chapter 12

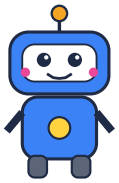


Learn

A **repeat until** loop keeps going until a condition becomes true. We do not need to know the number of repeats in advance.

Clocks and calendars are **cycles**: after 12 o'clock comes 1 again, and after Sunday comes Monday. 1 hour = 60 minutes; a week = 7 days.

repeat until	cycle	minute
loop until something is true	comes back to the start	60 make an hour



'Repeat until' means: keep going until something becomes true. Repeat until the clock shows 4:00, add 15 minutes!

Example

How many 15-minute steps from 3:00 to 4:00?

1. Start at 3:00. Repeat until the clock shows 4:00: add 15 minutes.
2. 3:15, 3:30, 3:45, 4:00.
3. The loop ran **4** times and then stopped.



★ *Coder fact!* Phones keep their clocks correct by checking time signals from the network.

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Try it!

Circle the right answer, or write it in the box.

1 Start at 2:00. Repeat until the clock shows 3:00: add 20 minutes. How many repeats?

2 What time does each clock show?



A B C

3 How many minutes are there in 2 hours 30 minutes?

4 1 March is a Monday. What day is 10 March?

a) Monday **b)** Tuesday **c)** Wednesday **d)** Sunday

5 An empty glass holds 500 ml. Repeat until the glass is full: add 100 ml. How many repeats?

Now you build it

Write a 'repeat until' program for brushing your teeth.

Start the timer | **Repeat until 2 minutes are over** | **brush 10 strokes** | **rinse**

Why is 'repeat until' better here than 'repeat 12 times'?

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★ Challenge

5 January is a Sunday. What day is 31 January?

✓ How did I do?

I can use a repeat-until loop.	I can read clocks to the quarter hour.	I can work out days with the week cycle.
		

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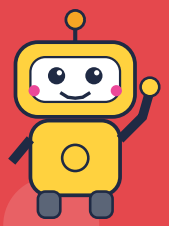
13

CHAPTER 13 · LEVEL B3

The Transport Museum

Coder skill: Decision trees: sorting by questions

Goes with NCERT maths, Class 4, chapter 13

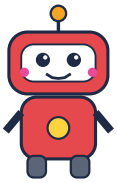


Learn

A **decision tree** sorts things by asking **yes/no questions**. Each answer takes you down a branch until you reach a group.

Good questions split things into useful groups. Some AI systems use decision trees to make choices.

decision tree	branch	classify
yes/no questions that sort	a path in the tree	put into the right group

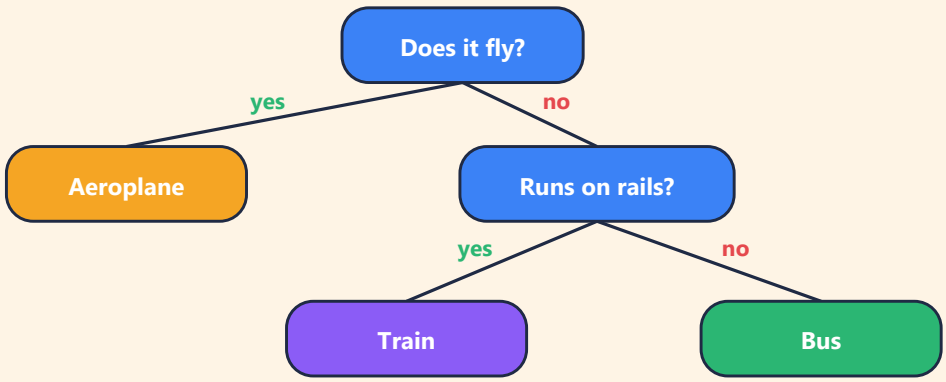


A decision tree asks yes/no questions, one after another. Some AI systems sort things this way too!

Example

Sorting the museum's vehicles

1. First question: *Does it fly?* Yes: **aeroplane**.
2. No: ask *Does it run on rails?* Yes: **train**. No: **bus**.
3. Two questions are enough to sort these three vehicles.



★ *Coder fact!* Some AI systems decide using decision trees: long chains of yes/no questions.

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Try it!

Circle the right answer, or write it in the box.

1 Use the tree. A tram does not fly but runs on rails. Which group is it in?

a) Aeroplane **b)** Train **c)** Bus

2 Which question would best split a car from a bicycle?

a) Does it have wheels? **b)** Does it have an engine?

c) Is it on the road?

3 The museum has 3 cars and 2 bicycles. How many wheels in all?

4 Which is the odd one out?

a) bus **b)** truck **c)** boat **d)** car

5 Sort into 'has an engine' and 'no engine': cycle, auto-rickshaw, bullock cart, scooter.

Now you build it

Design a decision tree two levels deep (two questions on every path) that sorts a fish, a parrot, a cow and a frog.

Question 1: ___ ? **if yes, ask: ___ ?** **if no, ask: ___ ?**

Test your tree with all four animals. Does each end up alone?

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★ Challenge

What is the fewest number of yes/no questions you need to find one vehicle out of 8, if each question splits the group in half?

✓ How did I do?

I can follow a decision tree.	I can ask good yes/no questions.	I can build my own tree.
		

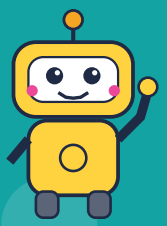
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CHAPTER 14 · LEVEL B2 review

14 Data Handling

Coder skill: Collecting and showing data

Goes with NCERT maths, Class 4, chapter 14

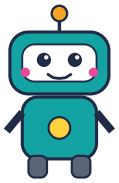


Learn

To answer a question like 'Which fruit does our class like most?', we **collect data**. Tally marks make counting quick: |||| with a line across means 5.

A **bar graph** shows the counts as bars. The tallest bar is the most popular. Computers and AI learn from data like this.

tally marks	bar graph	survey
lines for quick counting	bars that show counts	asking a question to many people

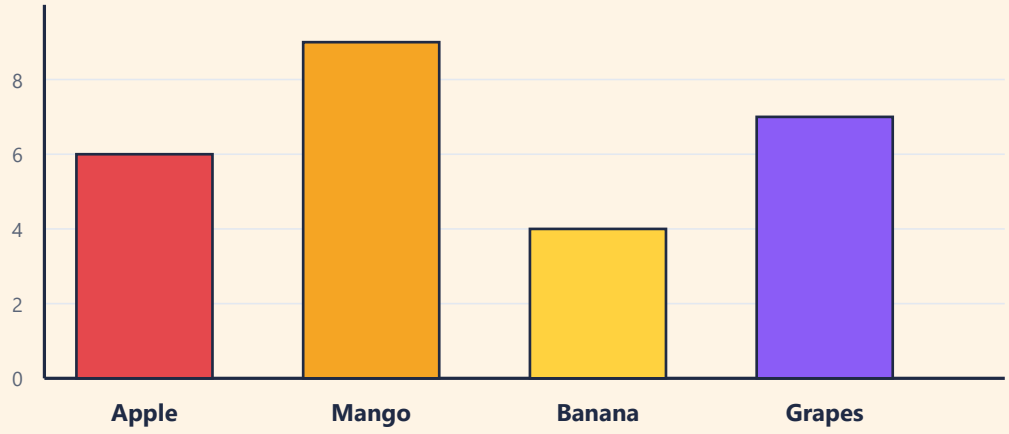


Data answers questions! I collect it with tally marks, count it, and show it in a bar graph so everyone can see the answer.

Example

Our class's favourite fruit

1. Ask everyone and make tally marks for each fruit.
2. Count the tallies: apple 6, mango 9, banana 4, grapes 7.
3. Draw a bar for each fruit. Mango has the tallest bar.



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Try it!

Circle the right answer, or write it in the box.

1. Use the bar graph above. How many children chose grapes?

2. Which fruit was chosen the least?

   a) apple **b)** mango **c)** banana **d)** grapes

3. How many children answered the survey in all?

4. How many more children chose grapes than banana?

5. Write 12 using tally marks.

Now you build it

Plan a class survey. Put the steps in order.

Draw a bar graph **Ask a question** **Count the tallies** **Make tally marks**

Carry out your survey and write one thing your graph shows.

Challenge

4 more children chose banana. Which fruit is now the least popular?

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✓ How did I do?

I can make tally marks.	I can read a bar graph.	I can answer questions with data.
		

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Answers

Many questions have more than one good answer. Accept any answer that follows the rule and can be explained.

1. Shapes Around Us

1. 6 faces, 12 edges, 8 corners
2. c) D
3. b) the 2 by 3 block
4. 5
5. a
Build it: Yes: the cross net folds into a cube; faces 1 and 3, 2 and 4 in the column, and the two side squares, are opposite
Challenge: 1 (the cube in the very middle)

2. Hide and Seek

1. Step, Splash, Splash, Step
2. jump, clap, jump
3. c) 4
4. b) blue
5. a) C2
Build it: Repeat 5 times: move forward; if on a sun tile then say 'Hot!' else say 'Cold!'
Challenge: 3 (tiles 3, 6 and 9)

3. Patterns Around Us

1. 27 and 41
2. Row 1 white, row 2 black, row 3 black
3. b) row 2, column 1 (row 2 and column 1 are now odd)
4. a) even
5. 32 (double each time)
Build it: Lay out a 4 x 4 grid; add check coins to make each row and column even; friend flips one coin; find the odd row and column
Challenge: No: the row stays even and two columns turn odd, so you know something changed but not exactly which coins

4. Thousands Around Us

1. 4, 0, 7, 2
2. 3,041; 3,140; 3,401; 3,410
3. b) 3
4. 4, 6, 9
5. 8,510
Build it: Yes: these three comparisons always sort three people
Challenge: No: 3, 2, 1 becomes 2, 3, 1 and then 2, 1, 3, which is not sorted, so the third comparator is needed

5. Sharing and Measuring

1. b) 3/8
2. a) one half
3. b) no: 15 is odd
4. c) Rs 6
5. 4
Build it: Not fair! (4, 3, 3 are not all the same)
Challenge: 15 (one quarter is 5, so three quarters is 15)

6. Measuring Length

1. 200 cm
2. b) square
3. 6 moves (right 4, up 2)
4. a) 2 m (200 cm)
5. b) down
Build it: Repeat 2 times: forward 3, turn right, forward 2, turn right; yes, it ends at the start
Challenge: No: it draws 3 sides of a square; Robo needs a 4th side to get back to the start

7. The Cleanest Village

1. b) both OFF
2. a) A ON, B OFF
3. 1 (press 2)
4. 3
5. after 2: ON, ON; after 1: OFF, ON
Build it: Open answer
Challenge: 2 presses: press 2 (both ON), then press 1 (A OFF)

8. Weigh It, Pour It

1. a) apple
2. b) two hands
3. 1,000
4. A, B, C
5. 2 litres
Build it: 1 kg: bike; 3 kg: truck; 2 kg: bike (2 is not more than 2)
Challenge: Fill 5; pour into 3 (2 left in 5); empty 3; pour the 2 into 3; fill 5; pour into 3 until full (1 goes); 4 litres stay in the 5 L jug

9. Equal Groups

1. 15
2. 24
3. a) 3 x 6 = 18
4. 30
5. 2 rows of 3 circles
Build it: Repeat 4 times: [repeat 5 times: plant a seed], start a new row; 20 seeds
Challenge: 6 claps and 3 stamps

10. Elephants, Tigers and Leopards

1. Asha blue, Ben red, Chitra green
2. 25
3. b) tiger
4. c) tiger
5. 300 kg
Build it: Open answer
Challenge: Leopard 1, deer 2, tiger 3, elephant 4

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11. Fun with Symmetry

1. A and H
2. c) F
3. c) 4
4. 4
5. b) left
Build it: Repeat 6 times: draw a petal, turn; the flower has lines of symmetry through the petals
Challenge: B2

12. Ticking Clocks and Turning Calendar

1. 3
2. A 4:15, B 7:45, C 10:30
3. 150
4. c) Wednesday
5. 5
Build it: We care about the time, not the number of repeats; the loop stops when 2 minutes are over
Challenge: Friday (26 days later = 3 full weeks + 5 days)

13. The Transport Museum

1. b) Train
2. b) Does it have an engine?
3. 16
4. c) boat (it does not run on roads)
5. Engine: auto-rickshaw, scooter. No engine: cycle, bullock cart
Build it: Example: Does it live in water? Yes: Does it live only in water? (yes fish, no frog). No: Can it fly? (yes parrot, no cow)
Challenge: 3 (8, then 4, then 2, then 1)

14. Data Handling

1. 7
2. c) banana
3. 26
4. 3
5. Two groups of five (each four lines with one across) and two more lines
Build it: Ask a question; make tally marks; count the tallies; draw a bar graph
Challenge: Apple (banana now has 8, apple has 6)

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Coder words

array	objects in rows and columns	Ch 9
bar graph	bars that show counts	Ch 14
branch	a path in the tree	Ch 13
centimetre	100 make 1 metre	Ch 6
classify	put into the right group	Ch 13
clue table	a grid to track clues	Ch 10
comparator	compares two numbers and orders them	Ch 4
condition	the question that is true or false	Ch 2
cycle	comes back to the start	Ch 12
data	facts we collect and count	Ch 10
decision tree	yes/no questions that sort	Ch 13
edge	where two faces meet	Ch 1
even	shares into pairs	Ch 3
face	a flat side of a solid	Ch 1
fraction	equal parts of a whole	Ch 5
heavier	weighs more	Ch 8
if-then-else	a choice between two actions	Ch 2
kilogram	1,000 grams	Ch 8
line of symmetry	the fold line	Ch 11
litre	a unit for liquids	Ch 8
logic	reasoning step by step	Ch 10
minute	60 make an hour	Ch 12
mirror image	a flipped copy	Ch 11
nested loop	a loop inside a loop	Ch 9
net	a flat shape that folds into a solid	Ch 1
odd	one left over	Ch 3
parity check	an even/odd check for mistakes	Ch 3
quarter	one of 4 equal parts	Ch 5
repeat until	loop until something is true	Ch 12
row	a line going across	Ch 9
shortest path	the route with fewest steps	Ch 6
sorting	putting in order	Ch 4
state	how things are right now	Ch 7
survey	asking a question to many people	Ch 14
symmetry	both halves match	Ch 11
tally marks	lines for quick counting	Ch 14
thousands	the 4th place from the right	Ch 4

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toggle	switch ON to OFF or OFF to ON	Ch 7
trace	follow step by step	Ch 7
turn	change the way Robo faces	Ch 6

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Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

up	down	left
up	down	left
right	Start	Repeat ___ times
right	Start	Repeat ___ times
If ___	else	When ___ happens
If ___	else	When ___ happens
add ___	take away ___	Stop
add ___	take away ___	Stop

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